Ref: HRSU/SDS/CNT/001 **Date: 09/10/2026 | Version 1.0**

SAFETY DATA SHEET: CALCIUM NITRATE TETRAHYDRATE

SECTION 1: PRODUCT AND COMPANY IDENTIFICATION

- **Product Name:** Calcium Nitrate Tetrahydrate
- **Other Names:** Nitric Acid, Calcium Salt, Tetrahydrate; Norwegian Saltpeter Tetrahydrate; Calcium Dinitrate Tetrahydrate
- **CAS No.:** 13477-34-4
- **EINECS EC Number:** 233-332-1
- **Molecular Formula:** $Ca(NO_3)_2 \cdot 4H_2O$
- **Molecular Weight:** 236.15 g/mol
- **Relevant Uses:** Industrial use only.
- **Supplier:**
  - **Entity Name:** HRSU Indore Pvt. Ltd.
  - **Registered Office:** UG-3, Abhishek Avenue, Slice-4, Scheme No. 78, Vijay Nagar, Indore (M.P.) 452010
  - **Manufacturing Unit:** Plot No. 62 MPIDC, Industrial Area Maksi, District Shajapur (M.P.) 465106
  - **Email:** info@hrsuindore.com
  - **Phone / Emergency Contact:** +91 9425000484

SECTION 2: HAZARDS IDENTIFICATION

- **GHS Classification:**
  - Oxidizing solids: Category 3
  - Acute toxicity, Oral: Category 5
  - Skin irritation: Category 3
  - Serious eye damage/eye irritation: Category 2B
- **GHS Label Elements:**
  - **Signal Word:** Warning
  - **Hazard Statements:**
  - H272: May intensify fire; oxidizer.
  - H303: May be harmful if swallowed.
  - H316: Causes mild skin irritation.
  - H320: Causes eye irritation.
  - **Precautionary Statements:**
  - P210: Keep away from heat/sparks/open flames/hot surfaces. No smoking.
  - P220: Keep/Store away from clothing/other combustible materials.
  - P280: Wear protective gloves/protective clothing/eye protection/face protection.
  - P305+P351+P338: IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
  - P312: Call a POISON CENTER or doctor/physician if you feel unwell.
  - P501: Dispose of contents/container to authorized agents.

HRSU Indore Pvt. Ltd.
Beyond Granules. The Purity of Powder.

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

- **Substance:** Calcium Nitrate Tetrahydrate, $Ca(NO_3)_2 \cdot 4H_2O$
- **Assay:** 99 – 103% (titrimetric)
- **CAS No.:** 13477-34-4
- **EINECS EC Number:** 233-332-1

SECTION 4: FIRST AID MEASURES

- **Inhalation:** Remove to fresh air. If not breathing, give artificial respiration. If breathing is difficult, give oxygen. Get medical attention.
- **Ingestion:** Never give anything by mouth to an unconscious person. Get medical attention.
- **Skin Contact:** Immediately flush skin with plenty of water for at least 15 minutes. Remove contaminated clothing. Get medical attention.
- **Eye Contact:** Immediately flush eyes with plenty of water for at least 15 minutes, lifting lower and upper eyelids occasionally. Get medical attention immediately.

SECTION 5: FIRE-FIGHTING MEASURES

- **Flammability:** Non-flammable, but it will support combustion.
- **Fire Extinguishing Media:** Use water spray or any means suitable for extinguishing the surrounding fire.
- **Special Information:** As an oxidizer, it may accelerate burning. In the event of a fire, wear full protective clothing and a self-contained breathing apparatus. Containers may explode in the heat of a fire.

SECTION 6: ACCIDENTAL RELEASE MEASURES

- **Personal Precautions:** Ventilate the area. Avoid breathing dust. Use personal protective equipment (gloves, safety glasses, etc.).
- **Environmental Precautions:** Do not let the product enter drains.
- **Cleanup Procedures:** Use appropriate tools to place the spilled solid in a waste disposal container. Clean the contaminated surface by spreading water and dispose of it according to local authority requirements.

SECTION 7: HANDLING AND STORAGE

- **Handling:** Wash thoroughly after handling. Minimize dust generation. Avoid contact with skin, eyes, and clothing.
- **Storage:** The product is hygroscopic and absorbs moisture from humid air. Store in a cool, dry, and ventilated area away from heat and sunlight, in a tightly closed container. Do not store with incompatible materials such as reducing agents, organic materials and combustible materials.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

- **Exposure Limits:** No specific occupational exposure limit values are established for this product.
- **Ventilation:** A system of local and/or general exhaust is recommended.
- **Personal Respirators:** For conditions where exposure to dust is apparent, a particulate respirator may be worn.
- **Skin Protection:** Wear protective gloves and clean body-covering clothing.
- **Eye Protection:** Use chemical safety goggles or a full-face shield where dusting is possible.

HRSU Indore Pvt. Ltd.
Beyond Granules. The Purity of Powder.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

- **Appearance:** Colourless to white crystals / crystalline powder.
- **Odor:** Odorless.
- **pH:** 5.0 – 7.0 (5% solution in water, 25 °C).
- **Melting Point:** 42 – 44 °C.
- **Solubility:** Freely soluble in water (typically 121 g in 100 g of water).
- **Density:** Approx. 1.86 g/cm^3 (typical).
- **Molecular Weight:** 236.15 g/mol.

SECTION 10: STABILITY AND REACTIVITY

- **Stability:** Stable under ordinary conditions of use and storage. Hygroscopic.
- **Hazardous Decomposition Products:** Emits toxic fumes (nitrogen oxides, calcium oxide) when heated to decomposition.
- **Incompatibility:** Reactive with reducing agents, organic materials and combustible materials.

SECTION 11: TOXICOLOGICAL INFORMATION

- **Toxicity to Animals:** LD50 [oral, rat]: 3900 mg/kg.
- **Carcinogenic Effects:** Not identified as a carcinogen by IARC.

SECTION 12: ECOLOGICAL INFORMATION

- **Toxicity to fish:** LC50 - Oncorhynchus mykiss (rainbow trout) - > 98.9 mg/l - 96 h.
- **PBT and vPvB assessment:** This substance is not considered to be persistent, bioaccumulative, or toxic (PBT).

SECTION 13: DISPOSAL CONSIDERATIONS

- Waste must be managed in an appropriate and approved waste disposal facility. Dispose of the container and unused contents in accordance with national, state, and local regulations.

SECTION 14: TRANSPORT INFORMATION

- **UN Number:** UN 1454
- **Proper Shipping Name:** Calcium Nitrate
- **Hazard Class:** 5.1 (Oxidizing Material)
- **Packing Group:** III
- **Applies to:** Road, rail, sea (IMDG) and air (IATA) transport.

SECTION 15: REGULATORY INFORMATION

- Classified under GHS as described in Section 2. The user should check and comply with all applicable national and local regulations for the storage, handling, transport and disposal of oxidizing substances.

SECTION 16: OTHER INFORMATION

- **Date of Issue / Version:** 09/10/2026 / 1.0
- **Disclaimer:** The information provided is believed to be accurate and represents the best information currently available. However, we make no warranty of merchantability or any other warranty, express or implied, with respect to such information.

HRSU Indore Pvt. Ltd.
Beyond Granules. The Purity of Powder.